

Sihan (Eric) Yan

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EDUCATION

University of Pennsylvania M.S. in Engineering in Data Science • <i>Relevant Coursework:</i> Advanced Machine Learning, Big Data Analytics, Time Series Forecasting, Database and Information Systems, Advanced Deep Learning, Natural Language Processing, Artificial Intelligence	GPA: 3.7/4.0 Philadelphia, PA Aug 2024 – May 2026
Arizona State University B.S. in Business Data Analytics B.S. in Supply Chain Management • <i>Relevant Coursework:</i> Business Data Analytics, Business Database Systems, Data Modeling, Enterprise Analytics	GPA: 4.0/4.0 Tempe, AZ Aug 2020 – May 2024 Mouer Award, Dean's List

SKILLS

Languages: Python, SQL, R, JavaScript, HTML/CSS
Analytical: Regression, Time Series, A/B Testing, Hypothesis Testing, PCA, Factor/Cluster Analysis
Machine Learning: KNN, Decision Trees, Random Forests, XGBoost, TensorFlow, PyTorch
Tools: Tableau (Certified), Power BI, Google Analytics, Salesforce, React, Node.js, Git
Databases: MySQL, PostgreSQL, MongoDB, Neo4j

PROFESSIONAL WORK EXPERIENCE

University of Pennsylvania Python, SQL, CRM <i>Data Analyst, Alumni Marketing</i>	Philadelphia, PA Aug 2024 – Present
- Conducted data collection through alumni surveys to analyze user demographics, career trajectories, and engagement behavior; derived insights that informed personalized outreach strategies, ultimately driving increased alumni involvement in departmental initiatives and startup investment opportunities. - Consolidated and cleaned 7k+ alumni records across 30 years into a master dataset using Python for data cleaning, joins, and aggregation, laying the foundation for automated data processing and improving efficiency by 40%. - Built 5 automated ETL pipelines integrating Qualtrics and HubSpot APIs, streamlining survey data processing and reducing manual workload by 80%.	
St. Vincent de Paul SQL, Python, Tableau <i>Data Scientist, Marketing Department</i>	Phoenix, AZ Aug 2023 – Apr 2024
- Collaborated with stakeholders to define campaign goals and business objectives, aligning efforts to boost revenue of donations through increased direct mail and Google Ads investment. - Performed financial planning and analysis to allocate a \$18,800 ad budget effectively, evaluating 5 advertising platforms to maximize ROAS and achieve a 40% increase in campaign effectiveness. - Developed 10 interactive Tableau dashboards with parameterized filtering and consolidated insights into Databricks, streamlining reporting workflows and providing a unified view of site performance—resulting in a 40% reduction in manual reporting time and improved data accessibility for cross-functional stakeholders. - Designed and automated ETL process in Databricks with SQL queries (e.g., window functions and joins) to seamlessly ingest ad and campaign data, eliminating manual input and increasing data pipeline speed by 30%, enabling quicker ad performance analysis and campaign adjustments. - Built churn prediction model using PyTorch, integrating categorical embeddings and numerical features into a custom MLP architecture; applied feature engineering, PCA, and stratified k-fold validation on behavioral logs to improve generalization, accuracy, and enabling targeted retention strategies that reduced churn rate by over 20%.	

PROJECT

News Source Classification NLP, Python	Aug 2024 – Dec 2024
- Built a 300k news headline dataset by scraping Fox and NBC with Selenium and BeautifulSoup, applying deduplication and text normalization to enable robust binary classification of news sources. - Built a NLP pipeline using BERT embeddings for news source classification, achieving 87% accuracy and outperforming baselines by 12% through optimized preprocessing and evaluation workflows.	
NBA Performance Analytics Platform Python, PostgreSQL	Sep 2024 – Dec 2024
- Cleaned and integrated 1.6M records from player stats, shot logs, and game outcomes; normalized 5 relational tables with referential integrity to support efficient season-level performance analysis. - Optimized SQL pipelines using CTEs and materialized views, reducing query latency from 22s to less 1s.	